Northern Bukidnon State College

GeoFare: A GIS-Based Habal-Habal Fare Computation and Management System Using Road-Network Distance in Manolo Fortich, Bukidnon

Presented to the

Faculty of the Institute for Computer Studies

Northern Bukidnon State College

In Partial Fulfillment

of the Requirements for the Degree

Bachelor of Science in Information Technology

By

Auggie John Fred G. Manginsay

Rian S. Cat-awan

Weniel A. Raña

July 2025

APPROVAL SHEET

This Capstone Proposal entitled: “**TITLE**” is prepared and submitted by **First name MI. Last name, First name MI. Last name, First name MI. Last name** and **First name MI. Last name** in partial fulfillment of the requirements for the degree of **Bachelor of Science in Information Technology** has been examined and is recommended for acceptance and approval.

FIRST NAME MI. LAST NAME

Adviser

PANEL OF EXAMINERS

Approved by the Committee on **Proposal Defense** with a grade of ______________ on July_____, 2025.

SHIELA MAE M OROOZCO, MIT

Chairman

FIRST NAME MI. LAST NAME

Member

FIRST NAME MI. LAST NAME

Member

Approved and accepted in partial fulfillment of the requirements for the degree Bachelor of Science in Information Technology.

SHIELA MAE M OROOZCO, MIT

Program Head

ACKNOWLEDGMENT

The Researchers

ABSTRACT

KEYWORDS:

TABLE OF CONTENTS

LIST OF FIGURES

LIST OF TABLES

CHAPTER 1

THE RATIONALE

1.1 Introduction

In rural and peri-urban municipalities across the Philippines, informal transport systems serve as a primary lifeline for communities lacking access to traditional public utility vehicles. In the municipality of Manolo Fortich, Bukidnon, motorcycle taxis locally referred to as habal-habal navigate complex, rugged terrains to connect remote agricultural barangays to major economic, healthcare, and educational centers along the Sayre Highway. Residents rely heavily on these riders to bridge the gap left by fixed-route transportation.

Despite their critical role, the motorcycle taxi sector operates under a severe operational vulnerability: arbitrary and unstructured fare pricing. Fare determination remains dependent on manually maintained printed matrices issued by the Local Government Unit (LGU), or worse, unguided estimations. This lack of standardization leads to constant fare disputes, passenger exploitation, and income instability for riders who cannot accurately account for volatile fuel prices or demanding route elevations.

Early academic interventions at NBSC attempted to address local transport issues, most notably through *Transmobil Ally: A Transportation Mobile Ally* by Nailes and Lagmay (2020). However, the application suffered from foundational technical limitations: it lacked availability outside limited terminal boundaries, possessed no connection to external mapping frameworks, failed to integrate GPS route telemetry, and completely omitted road-network distance calculation.

To address these limitations, GeoFare introduces an automated Geographic Information System (GIS)-based architecture for habal-habal fare computation. Rather than relying on inaccurate straight-line (Euclidean) distance estimations, the system utilizes road-network distance analysis to compute the shortest navigable route between selected locations. By calculating travel distances using actual road networks and applying the LGU-approved fare matrix, GeoFare automates fare computation, providing a more transparent, accurate, and consistent approach to fare determination in Manolo Fortich's habal-habal transportation system.

Review of Related Literature

A. Local Literature

The Baseline Local Technology Gap. Nailes and Lagmay (2020) in their NBSC capstone project, *Transmobil Ally: A Transportation Mobile Ally*, established the initial framework for local ticketing and driver alignment. The proponents explicitly noted that the application did not use Google Maps or a map model to locate the driver's location and lacked GPS integration to track the driver's route. They recommended redeveloping the system to embed real-time tracking and automated fare circulation, which directly forms the technical foundation of GeoFare.

Ride-Matching and Regulatory Transparency. Nesnia (2026) developed *TRIGO: A Smart Tricycle Ride-Matching and Fare Management System* at Davao Central College. This study emphasizes that informal paratransit modes in rural environments suffer from a lack of accountability and service clarity. By automating fare verification via digital ride-matching, TRIGO demonstrated how targeted software solutions enhance service transparency and policy compliance within local communities.

Automated Collection Architecture. Espanol et al. (2025) presented *PAYSADA: An Architectural Design for an Automated Fare Collection and Tracking System for Jeepneys*. Their work outlines the architectural design required to handle mobile transit transactions smoothly. It highlights that public utility systems must rely on structured backend synchronization to track vehicle locations accurately while preventing database lockups during concurrent fare requests.

Mathematical Driver Frameworks. Tica-a et al. (2025) analyzed informal pricing mechanisms in *Calculating the Route: Uncovering the Mathematical Strategies Employed by Buyagan-Baguio Route Jeepney Drivers*. The researchers found that drivers intuitively create unstructured pricing rules based on terrain friction, passenger loads, and road quality. This underscores the need to convert these informal human calculations into concrete digital algorithms within a standardized software engine.

Geospatial Processing in the Philippine Context. Narboneta and Teknomo (2020) demonstrated the utility of combining OpenTripPlanner, OpenStreetMap (OSM), and GTFS for transport analysis within the Philippines. Their findings confirm that open-source mapping databases provide sufficiently granular road network layers to handle complex transit calculations in municipal sectors where formal commercial mapping data is scarce or restricted.

1.2 Statement of the Problem

1.3 Objectives of the Study

This study intends to …., the study aims:

- To
- To

• To

• To

• To

1.4 Scope and Limitations of the Study

1.5 Significance of the Study

The following are the beneficiaries of the system:

• **To Northern Bukidnon State College (NBSC) -**

• **To the users -**

• **To the researchers –**

• **Future Researchers –**

1.6 Definition of Terms

(SAMPLE ONLY)

Machine Learning. It is a discipline of artificial intelligence (AI) that provides machines with the ability to automatically learn from data and past experiences while identifying patterns to make predictions with minimal human intervention.

Image Processing. It is a method to convert an image into digital form and perform some operations on it, in order to get an enhanced image or extract some useful information from it. It is a type of signal dispensation in which input is image, like video frame or photograph and output may be the image or characteristics associated with that image.

Image Classification. In conventional machine learning, it is defined as a technique of predicting the class membership of testing the data instances merely on the basis of the knowledge of the class memberships of the training data. Here the training data is the known data that enables the machine learning algorithm to learn and then perform the task classification, and the testing data is considered the unknown data that the machine learning classifier or algorithm needs to correctly classify.

Dataset. It is a collection of data that is treated as a single unit by a computer. This means that a dataset contains a lot of separate pieces of data but can be used to train an algorithm with the goal of finding predictable patterns inside the whole dataset.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Related Literature and Studies

(SAMPEL ONLY)

For further understanding of the study, the researchers made use of related published materials that are essential in broadening the knowledge of the proponents. These will also serve as guide in achieving their target objectives by referencing on other related studies and making improvements as possible.

A solution to enable the society to automate and optimize waste management processes using Internet of Things (IOT) technologies was developed at Sri Lanka Institute of Information Technology. When the waste bins are full or if there is an unusual condition inside the bins, such as high temperature or high humidity, the proposed solution will notify the user and other authorities with Alert Notification Sub System. Moreover, the authors used sound and light indicators to determine bin status with dashboard (Kumari et al. 2018).

A university in Pune, India, developed a smart dustbin that automatically separates dry and wet waste. The smart waste-bin design is based on both software and hardware implementation. Arduinos, sensors, servo motors, and batteries are among the hardware components used. The smart waste-bin has two main steps: motion detection and dry-wet waste sorting. The project was successful in meeting the lid opening scenario and segregating dry and wet waste. To avoid human contact or interference, motion detection is used. The researchers suggested using solar panels to make the Smart Waste Bin more energy efficient. Similarly, the proposed bin currently has two compartments in a single container; however, the bin design is recommended for redesign so that different types of waste can be separated into different containers. Furthermore, suggesting AI features for the system to make waste management even easier (Mahendrakar et al. 2020).

2.2 Synthesis of the Reviewed Related Literature and Studies

(SAMPLE ONLY)

The above collection of related literature provided information that the current study, while similarities were seen with other systems, proponents placed greater interest in the research gaps found to produce innovative ideas. Thus, giving the proposed system an edge among the existing studies related to automated waste segregation.

The above studies influenced the realization of the key features of the GULP (Great User-oriented Litter Placer): Solar-Powered Smart Garbage Segregation Bins with SMS Notification and Machine Learning Image Processing. There are supporting studies that stated that solar panels should be included in the project to make it more energy efficient. Several studies with IoT-reliant smart bins, also offered a different perspective by recommending the use of image processing for more accurate waste classification. They asserted that the IoT driven projects' constant need for internet connectivity in order to make data readily available to the control system may be a challenge in countries where data on-the-go services are not well established. A GSM module can be used instead to monitor bin status via SMS alerts. Furthermore, additional studies made a compelling case for using incentives to encourage people to participate in the common goal of proper waste segregation.

CHAPTER 3

METHODOLOGY

3.1 Research Design

3.1.1 Type of Research

(SAMPLE ONLY)

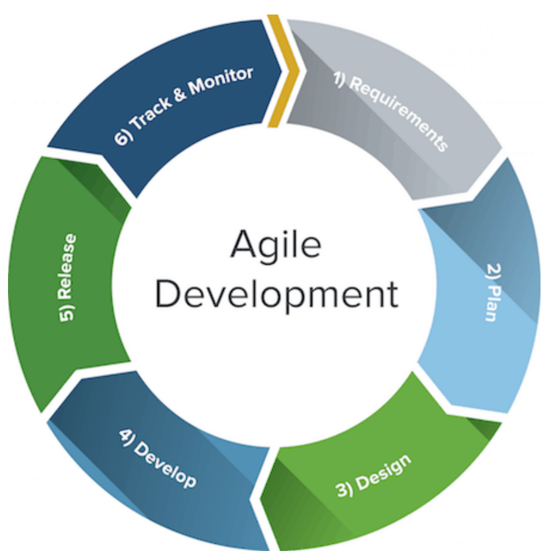
This research study used the developmental and quantitative research designs. Developmental research is described as "the systematic study of designing, developing and evaluating instructional programs, processes, and product that must meet the criteria of internal consistency and effectiveness"(Richey 1994). Developmental research seeks to generate knowledge based on the data systematically derived from practice. It is a pragmatic type of research that provides a means to test "theory" that has only been hypothesized and to validate practice that has largely been perpetuated through unchallenged tradition. The proponents made use of Type 1 developmental research, also labeled "system-based evaluation," in which the product development process is described and analyzed and the final product is evaluated. The outcome of this type of developmental research is derived from the lesson learned from developing specific product and analyzing the conditions that facilitate their use.

On the other hand, quantitative research involves the utilization and analysis of numerical data using specific statistical techniques to answer questions like who, how much, what, where, when, how many, and how. It also describes the methods of explaining an issue or phenomenon through gathering data in numerical form (Apuke 2017). In this paper, the researchers used a quantitative research design to analyze ISO/IEC 25010 for product quality evaluation and it is valued according to the Likert scale system.

3.2 Research Design Approach

(SAMPLE ONLY)

In this study, the researchers used an Agile Development approach since this method can help teams manage work more efficiently and do the work more effectively while delivering the highest quality product within the constraints of the budget. This method follows six fundamental phases, namely: requirements analysis, planning, design, coding, unit testing and acceptance from the client.



Source: (Anon 2022)

Figure 1. **Agile Development**

Requirements Analysis

Before the proponents began creating the proposed project, they first generated the initial documentation, which includes a description of the project's needs. This phase entails reviewing and analyzing the requirements, such as the main variables of the study. By conducting a discussion, many ideas and concepts on how the project's flow will work out are collected.

Planning

The Gantt chart clearly specified the sequence of activities that were undertaken during the development of the study. It began with information gathering, which is the most vital activity that a project must perform in order to clearly identify the purpose and benefits of the project, and ended with the submission of the hardbound manuscript.

Design

In this phase, there were two approaches taken in designing the GULP smart bin: one was *visual design,* and the other was the *architectural structure* of the application. During the first iteration, the project manager gathered the rest of the team and presented the requirements produced during the previous stage. The entire team then discussed how to approach these goals and proposed tools needed to achieve the best results. The System Design Architecture, Block Diagram, and Use Case Diagram were some of the tools used that guided the developers in realizing the architectural structure of the project.

The project designer also constructed a rough mock-up of the user interface. It is critical to evaluate user interface and user experience when they are done correctly and, more importantly, when they are not. In subsequent iterations, the fundamental design is refined to accommodate the new features.

Implementation, coding or development ***(Do not write this part, yet)***

This phase includes the actual development that occurs after designing a prototype. In developing the application, researchers chose the Thonny Python IDE and Arduino IDE while using Python and Arduino programming language (framework based on the C++ language) as the main programming languages that allowed the researchers to inscribe computer commands. Furthermore, Lobe.ai was used as a tool in developing machine-learning models for image classification.

Testing ***(Do not write this part, yet)***

This phase involves the product's development to test its functionality, and the researchers repeat the process until no errors are found. Table 2 shows the waste types tested which are few of the samples found in the dataset. The researchers conducted tests on various types of waste to determine whether the garbage was correctly sorted.

Deployment ***(Do not write this part, yet)***

Finally, the researchers present their output system to the proposed organization for approval through presentation and system demonstration. A PowerPoint presentation was used for the preparatory and concise flow of the system while system demonstration was done for genuine presentation. The researchers also used ISO/IEC 25010 to evaluate the system's quality. The ISO/IEC 25010 evaluation functions to identify the quality features and properties of the product, as well as a reliable evaluation tool that is appropriate for the study.

3.3 Hardware and Software Specification

The following are the hardware and software specifications of the project. These were requirements for the implementation and design of the application.

3.3.1 Hardware Specification for Development

The table shown below reflects the hardware components of the project during the development phase of the researchers.

Table 1. **Hardware Development Specification**

HARDWARE	**SPECIFICATION**
Lenovo ideapad 330s 14ikb	Processor: Intel(R) Core(TM) i3-7020U CPU @ 2.30GHz 2.30 GHz Installed RAM: 8.00 GB System type: 64-bit operating system, x64-based processor ***Atleast or Higher than the above-mentioned specification*

3.3.2 Software Specification for Development

Table 2 discussed the software specification used to develop the system.

Table 2. **Software Development Specification**

SOFTWARE	SPECIFICATION
Thonny Python IDE	Python Version: Python 3.10 Minimum Tk Version: Requires Tk 8.6 File Size: Approximately 23.48 MB License: Free and open-source Supported Platforms: Windows, macOS, and Linux ***Atleast or Higher than the above-mentioned specification*

3.3.3 Hardware Specification for Deployment ***(Do not write this part, yet)***

Table 3 discussed the hardware specification used during the project's deployment stage.

Table 3. **Hardware Deployment Specification**

HARDWARE	SPECIFICATION
Computer Desktop	Processor: Intel(R) Core(TM) i3-7020U CPU @ 2.30GHz 2.30 GHz Installed RAM: 8.00 GB System type: 64-bit operating system, x64-based processor ***Atleast or Higher than the above-mentioned specification*

3.3.4 Software Specification for Deployment *(Do not write this part, yet)*

Table 4 discussed the hardware specification used during the project's deployment stage.

Table 4. **Hardware Deployment Specification**

HARDWARE	SPECIFICATION
Thonny Python IDE	Python Version: Python 3.10 Minimum Tk Version: Requires Tk 8.6 File Size: Approximately 23.48 MB License: Free and open-source Supported Platforms: Windows, macOS, and Linux ***Atleast or Higher than the above-mentioned specification*

3.4 System Design Architecture

The diagram below presents the system design architecture of GULP (Great User-oriented Litter Placer): Solar-Powered Smart Garbage Segregation Bins with SMS Notification and Machine Learning Image Processing which describes the operational mode on how it works.

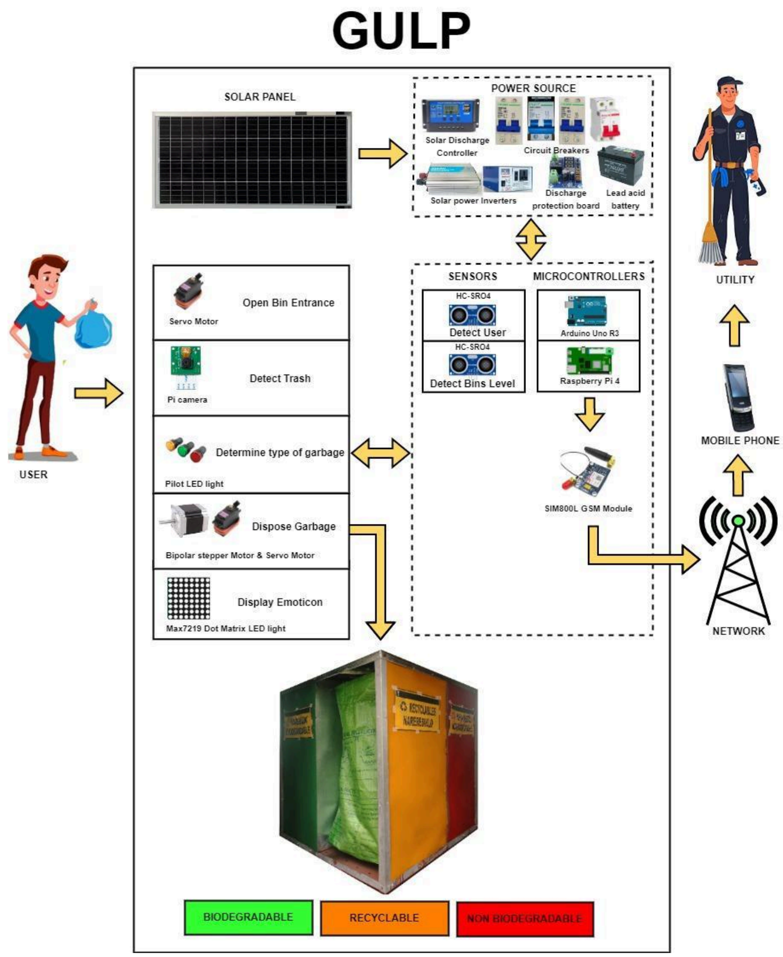


Figure 2. **System Design Architecture of GULP**

As shown in Figure 2, the architecture organizes the main components needed to implement the application and the processes that occur during use. The physical machine is comprised of various hardware components integrated for a specific use. The hardware phase is divided into two sub-phases: 1) Front End (application interface the regular user would interact with, and 2) Back End (parts of the application and their code that allow it to operate and that cannot be accessed by the regular user like the above diagram with external components enclosed in broken lines).

3.5 System Diagrams

Use-Case Diagram

The proponents used a Use-Case Diagram to describe what the application does and how the operators use it.

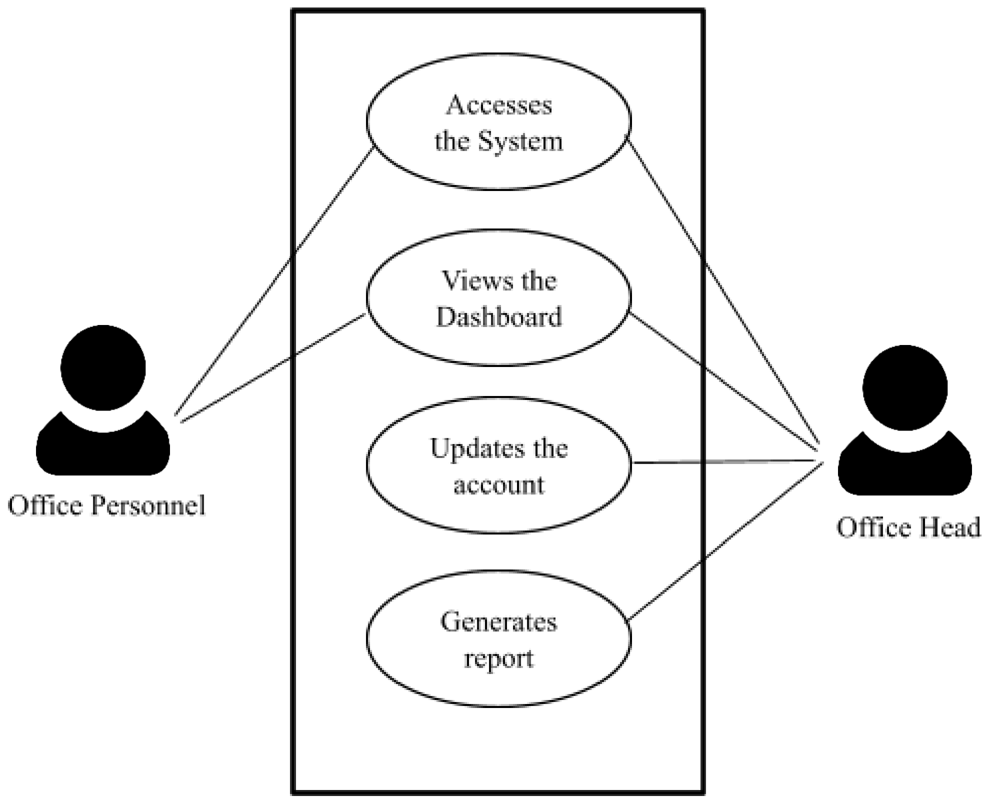


Figure 3. **Use-Case Diagram of**

(Explanation here)

Entity Relationship Diagram ***(Do not write this part, yet)***

Figure 4. **Entity Relationship Diagram of**

(Explanation here)

3.6 System Evaluation

3.6.1 Evaluation Tool

The performance of the application is assessed to see whether it complies with ISO/IEC 25010 Standards.

Table 6. **Likert Scale Table**

SCALE	PARAMETERS	VERBAL INTERPRETATION
5	4.20 – 5.00	Excellent (E)
4	3.40 – 4.19	Very Good (VG)
3	2.60 – 3.39	Good (G)
2	1.80 – 2.59	Fair (F)
1	1.00 – 1.79	Poor (P)

A Likert scale is a method for converting quantitative values into qualitative values in order to perform the best statistical analysis. It is commonly used to assess attitudes, knowledge, perceptions, values, and behavioral changes. A Likert-type scale consists of a series of statements from which respondents can select in order to rate their responses to evaluative questions. The researchers used the Likert Scale to interpret the result and overall average of the ISO/IEC 25010 evaluation.

Formula:

$$\frac{Total\ Average}{numbers\ of\ questioned\ item} = Overall\ Average$$

Table 7. **ISO/IEC 25010 Survey Test**

CHARACTERISTICS	SUB-CHARACTERISTICS	QUESTION	POOR 1	FAIR 2	GOOD 3	VERY GOOD 4	EXCELLENT 5
Functionality Suitability	Completeness	Does the system have complete features?					
	Correctness	Does the system provide correct information and function appropriately when used?					
	Appropriateness	Are the system's functionality and features appropriate to use?					
Performance Efficiency	Time-Behavior	Does the system perform the task in a timely manner?					
	Capability	Does the system meet the expected requirements?					
	Resource Utilization	Does the amount and type of resources of the system when performing functions meet requirements?					
Compatibility	Co-Existence	Does the system perform functions without affecting other records or components?					
	Interoperability	Does the system allow one to exchange information from one device to another and use that information?					
Usability	Appropriateness Recognizability	Is the system appropriate for your needs?					
	Learnability	Is the system easy to learn?					
	Operability	Is the system easy to use?					
	Use Error Protection	Does the system prevent users from making errors?					
	User Interface Aesthetics	Is the system design appropriate and user-friendly?					
	Accessibility	Is it possible for people with a diverse set of characteristics and abilities to use the system?					
Reliability	Maturity	Does the system meet the needs for reliability under normal operation?					
	Availability	Is the system reliable and convenient to use when needed?					
	Fault Tolerance	Does the system still function despite encountering some errors when used?					
	Recoverability	Can the system recover data in the event of a failure?					
Security	Confidentiality	Does the system protect user information and only allow authorized people to use it?					
	Integrity	Does the system prevent unauthorized access?					
	Non-Repudiation	Does the system provide information that an action has taken place?					
	Accountability	Can the system track entity actions?					
	Authenticity	Does the system require authentication that a user is really the authorized person accessing the system code?					
Maintainability	Modularity	Do the changes in the system component not affect other components?					
	Reusability	Can the system be used by different departments and agencies?					
	Analyzability	Does the system provide analytical reports such as errors and records?					
	Modifiability	Can the system be easily modified without affecting other components?					
	Testability	Does the system effectively and efficiently meet the test criteria?					
Portability	Adaptability	Does the system have the capacity to adapt to changes?					
	Installability	Can the system be easily installed and uninstalled?					
	Replaceability	Can the system be easily replaced by a new system in the same environment?					

3.6.2 Distribution of Respondents

Table 8. **Distribution of Respondents**

Respondents	No.	Percentage
Regular Users	22	73.33%
Utility Personnel	7	23.33%
Head of Health & Sanitation	1	3.33%
Total:	30	100%

Table 8 shows the total number of respondents. Twenty-two (22) regular users, seven (7) utility personnel, and the Head of Health and Sanitation. In total, there were thirty (30) individuals who evaluated the application.